# Communication Lower Bounds for Multi-Party Graph Traversal

## Quarter 2 Project Report


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### Abstract

A large number of computational problems can be modelled as communication games. For example, a graph traversal problem can be modelled as follows: Two players, Alice and Bob, have subgraphs of a known graph $G$, and wish to find a path from vertex $v$ to $w$ using only vertices and edges in either player's subgraphs. Communication complexity studies such situations by creating asymptotic bounds on the minimum number of communication necessary to complete these tasks. In this report, we prove communication lower and upper bounds for some graph traversal problems, such as the example above. We also survey how such bounds can be used by algorithm, system, and data structure designers.


Website: https://rybplayer.github.io/capstone/
Repository: https://github.com/rybplayer/DSCCapstone

# 1 Introduction

## 1.1 Motivating Communication Complexity

As networks and data grow in complexity and size, it will become increasingly difficult for individual machines to compute functions on such data. For example, suppose we wish to fly from San Diego to Santorini with the least budget. There is no single airline that has both flights out of San Diego and into Santorini. In other words, we will need to consult the flight database of various airlines to construct such a route. Does this mean that we have to search through every possible combination of flights and airlines? That is, do we need to ask every airline provider to communicate their entire schedule of flights to us? Are there techniques that we can employ to reduce the amount of flights we have to search through?

The amount of communication necessary to solve such problems is the primary objective of communication complexity. By studying communication complexity, we find lower and upper bounds on the minimum amount of communication necessary to solve such problems. In consequence, these bounds can inform on whether certain algorithms are optimal, or if it is possible to optimize them further. These bounds are increasingly relevant in a world full of distributed systems and data.

## 1.2 Communication Complexity Basics

Communication complexity studies the communication requirements to solve problems where inputs are distributed among at least 2 parties each with unbounded computing power. An example 2 party problem is typically stated as follows:

> There are two players, Alice and Bob, who respectively have a $n$-bit string, $a$ and $b$, that may or may not be the same. The goal is to compute a function, $f(a,b)$, that depends on both $a$ and $b$.

Under the two party model, it is always possible to compute $f(a,b)$ by sending all of $a$ to $b$, since we assume each party has unbounded computing power. This approach of sending the entire input from one party to another is what is called the *trivial solution*. We say problems are *maximally hard* or *maximally complicated* when we cannot do asymptotically better than the *trivial solution*.

We call such problems maximally hard because it has a deterministic communication complexity of $\Theta(n)$ bits. We call it *deterministic* as we do not allow error. Thus in maximally hard problems, Alice cannot do asymptotically better than sending the $n$ bits of $a$ to Bob and the same for Bob. An example of a maximally hard problem is 2-player equality (EQ):

**Claim 1.** *Let $a, b \in \{0,1\}^n$. That is, $a$ and $b$ are $n$-bit strings. Define equality as*

$$\mathsf{EQ}(a,b) = \begin{cases} 1 & a = b, \\ 0 & a \neq b. \end{cases} \tag{1}$$

*The deterministic communication complexity of two party equality, denoted $D(\mathsf{EQ})$, is $\Omega(n)$.*

*Proof.* We can represent the universe of all possible $n$ bit inputs to the two parties using a $2^n \times 2^n$ boolean matrix $M$ indexed by $a$ and $b$. Then, let $M_{a,b} = 1$ if $a = b$ and $M_{a,b} = 0$ otherwise.

Therefore, $M$ is the identity matrix. It is a well-known fact that the deterministic communication complexity of a problem is at least the $\log_2$ rank of its boolean matrix (Rao and Yehudayoff 2020). Therefore, since equality's matrix $M$ is the $2^n \times 2^n$ identity matrix with rank $2^n$, we have $D(\mathsf{EQ}) = \Omega(\log_2 2^n) = \Omega(n)$. □

$$M_{b,a} = \begin{array}{c|cccc} b\backslash a & 00 & 01 & 10 & 11 \\ \hline 00 & 1 & 0 & 0 & 0 \\ 01 & 0 & 1 & 0 & 0 \\ 10 & 0 & 0 & 1 & 0 \\ 11 & 0 & 0 & 0 & 1 \end{array}$$

Figure 1: Communication matrix for the equality function on two-bit inputs. The entry at column $a$ and row $b$ is 1 when $a = b$ and 0 otherwise.

As we will soon see, problems like equality are surprisingly useful for proving communication lower bounds for other, more complicated problems. That is, we say problem $A$ is *at least as hard* as problem $B$ if the communication complexity of $A$ is at least the communication complexity of $B$.

## 1.3 The Graph Traversal Problem

In this report, we study the problem of finding a path in a graph which is jointly owned by various players. This is an abstract version of the motivating problem in **??**. We choose to work in this abstract version so that that any bounds we prove will be applicable to a wide range of problems.

Informally, the graph traversal problem presents a graph, where players can own vertices. There is a start and end vertex. The player wish to know whether there exists a path from the start to the end through only vertices and edges owned by some player. Though each player knows what the graph looks like, they do not know who owns vertices and edges besides their own.

Formally, the graph traversal problem ($\mathsf{TRAV}$) is defined as follows:

**Definition 1.** *Let $G = (V, E)$ be an unweighted, undirected graph, where $V$ is the set of all nodes and $E$ is the set of edges. Each player $i$ owns a vertex subset $V_i \subseteq V$ of the graph, as well as all edges $E_i = \{e = (u, v) \in E \mid \{u, v\} \cap V_i \neq \emptyset\}$ connected to these vertices. Thus each player can be said owns a subgraph $G_i = (V_i, E_i)$ of $G$. Note that multiple players can own the same vertices and edges, and some vertices and edges may have no owner whatsoever.*

*Therefore, player inputs are solely determined by the vertices they own. Let the graph $G$ have $|V| = n$ vertices. Then each player's subgraph can be represented by $n$-bit strings $x$ and $y$, where 1 at index $j$ indicates the player owns vertex $j$, and 0 means the player does not own $j$. We may use $x, y$ and $G_1, G_2$ interchangeably, as they encode the same information. Given a fixed $G$, start vertex $v$, and end vertex $w$, we can an instance of two player* $\mathsf{TRAV}_2$ *as follows:*

$$\mathsf{TRAV}_2(x, y) : \{0,1\}^n \times \{0,1\}^n \to \{0,1\} \tag{2}$$

$$\mathsf{TRAV}_2(x, y) = \begin{cases} 1 & \exists \textit{ path from } v \textit{ to } w \textit{ through only vertices and edges in } G_1 \cup G_2. \\ 0 & \textit{There is no such path.} \end{cases} \tag{3}$$

The above definition generalizes to $\mathsf{TRAV}_k$, the traversal problem with $k$ players. The following figures illustrate what $\mathsf{TRAV}_3$ may look like.

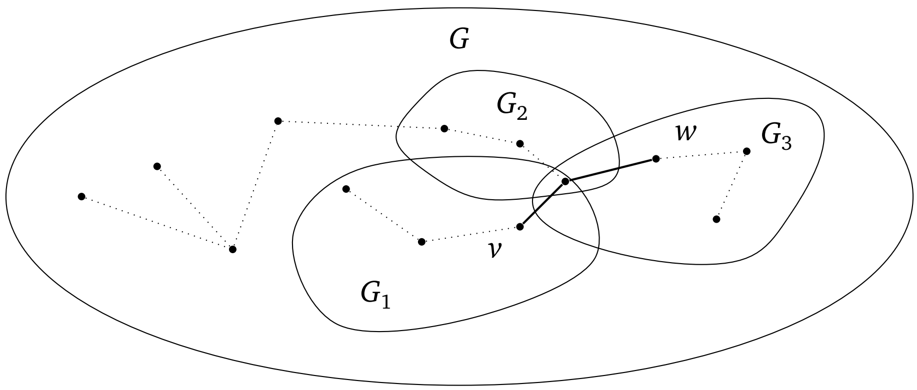


Figure 2: Example graph where $\mathsf{TRAV}(x, y) = 1$

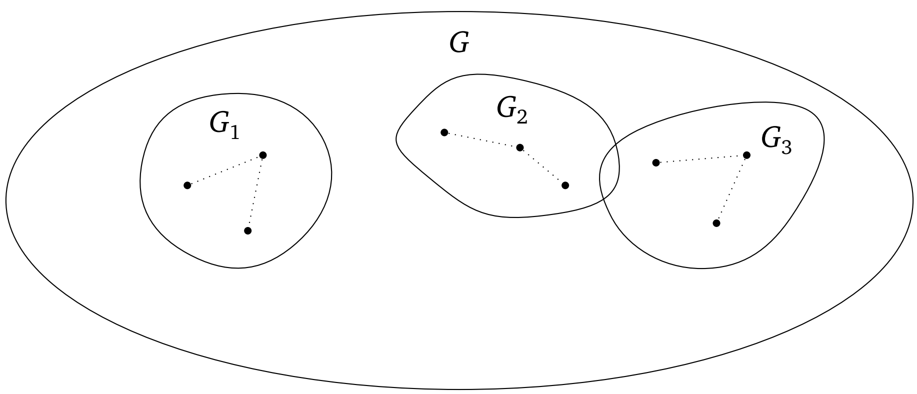


Figure 3: Example where $\mathsf{TRAV}(x, y) = 0$

# 2 The Two-Party Traversal Problem

## 2.1 Deterministic Two-party Bounds

We begin by considering the case with only two players and thus two subgraphs ($k = 2$). Let these players be named Alice and Bob, with subgraphs $G_A$ and $G_B$ respectively. Denote this problem by $\mathsf{TRAV}_2$, i.e. the traversal problem for 2 players, where $|V(G)| = n$ and $|E(G)| = O(n^2)$. We claim that $\mathsf{TRAV}_2$ is at least as hard as two player equality:

**Claim 2.** *Recall that equality* $\mathsf{EQ}$ *has inputs* $x, y \in \{0,1\}^n$*, and* $\mathsf{EQ}(x, y) = 1$ *iff* $x = y$*. Then* $D(\mathsf{TRAV}_2) \geq D(\mathsf{EQ})$*, and more generally* $\mathsf{TRAV}_2$ *is at least as as hard as* $\mathsf{EQ}$*.*

*Proof.* The high-level idea of this proof is to construct an equality gadget that lets us embed $\mathsf{EQ}$ into $\mathsf{TRAV}_2$. That is, given $x$ and $y$ as inputs of $\mathsf{EQ}$, we can convert these into subgraphs $G_1$ and $G_2$ inputs of $\mathsf{TRAV}$. We convert this in such a way that $\mathsf{TRAV}(G_1, G_2) = 1$ if and only if $\mathsf{EQ}(x, y) = 1$. Then $\mathsf{TRAV}$ cannot use less communication than the optimal $\mathsf{EQ}$ protocol, else we get a contradiction.

**Equality gadget.** We construct a gadget that enforces equality at a single bit position $i$ of $x$ and $y$. The gadget is shown in Figure **??**, and consists of two parallel branches:

1. The top branch corresponds to the assignment $x_i = y_i = 1$,
2. The bottom branch corresponds to the assignment $x_i = y_i = 0$.

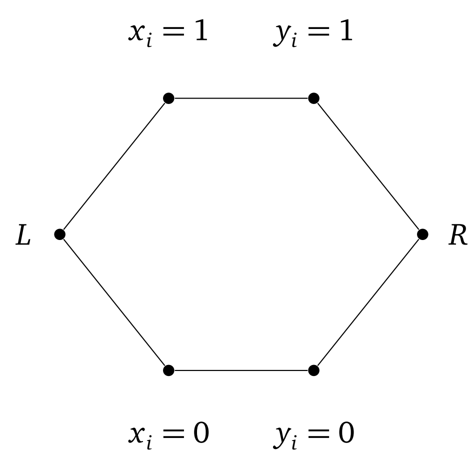


Figure 4: Gadget for proving equality. A path only exists if $x_i = y_i$ for either branch.

Add $L$ and $R$ into $G_A$ and $G_B$. Then, add $x_i = 1$ if this is true to $G_A$, else add $x_i = 0$. Do the same for Bob. Then a path from $L$ to $R$ exists if and only if Alice and Bob have vertices on the same branch, which holds exactly when $x_i = y_i$.

**Reduction to our graph problem.** Construct a graph of $n$ equality gadgets in a chain, as in Figure **??**. Construct subgraph $G_A$ as follows: Include all vertices at the start and end of each equality gadget (i.e. all the $L$ and $R$ above). Also, for every $i \in [n]$, give the vertex labelled $x_i = 1$ if this is true to $G_A$, else $x_i = 0$. Repeat for $G_B$.

In the resulting graph, set the leftmost vertex as the start vertex $v$, and the rightmost vertex as the end vertex $w$. Then there exists a path from $v$ to $w$ if and only if all gadgets are traversable, which occurs if and only if $x_i = y_i$ for all $i$, and therefore if and only if $x = y$.

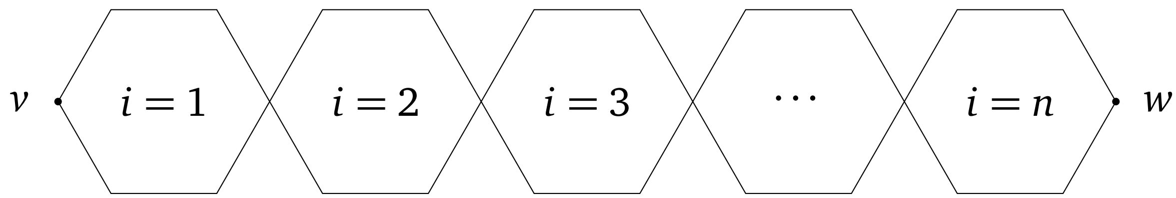


Figure 5: $n$ equality gadgets chained together.

We conclude by noting that the resulting graph $G$ has $\Theta(n)$ vertices and $\Theta(n)$ edges. As $|x| = |y| = \Theta(n)$, this completes the reduction and $\mathsf{TRAV}_2$ is at least as hard as $\mathsf{EQ}$. □

The trivial protocol where Alice sends all of $G_A$ to Bob has communication complexity $\Theta(n)$. This gives $D(\mathsf{TRAV}_2) = O(n)$. Together with Claim **??**, this gives $D(\mathsf{TRAV}_2) = \Theta(n)$. Thus $\mathsf{TRAV}_2$ is deterministically maximally hard.

## 2.2 Randomized Two-Party Bounds

Above, we note that the problem is *deterministically maximally hard*. However, allowing for some error may allow us to produce a randomized algorithm which much better cost.

Consider the *public coin randomness* model. In this model, both parties have access to the same random bit-string in addition to the problem description and input. This random bit string can be as long as we want. We also allow our output to error, that is, be incorrect, some bounded constant percent of the time, and strictly less than half the time. It can be shown that using the public coin randomness model, equality—which is Deterministically maximally hard—can be reduced to being $O(1)$. We denote this by $R(\mathsf{EQ}) = O(1)$.

Unfortunately, for $\mathsf{TRAV}_2$ we find that such a bound is unattainable. To prove this, we introduce the disjointness problem $\mathsf{DISJ}$, which has public coin randomized communication complexity $R(\mathsf{DISJ}) = \Theta(n)$. As we show, $\mathsf{DISJ}$ can be embedded into $\mathsf{TRAV}_2$.

**Claim 3.** *Define the disjointness problem as follows: Let $x, y \in \{0,1\}^n$ be n-bit strings representing subsets of $[n]$. Then, for all $i \in [n]$, if $x_i = 1$, we say $i$ is in Alice's subset $S_A \subseteq [n]$, and if $x_i = 0$ it is not. Similarly for $y$ and Bob. Then*

$$\mathsf{DISJ}(x, y) : \{0,1\}^n \times \{0,1\}^n \to \{0,1\} \tag{4}$$

$$\mathsf{DISJ}(x, y) = \begin{cases} 1 & S_A \cap S_B = \emptyset, \\ 0 & S_A \cap S_B \neq \emptyset. \end{cases} \tag{5}$$

*It is well-known that $D(\mathsf{DISJ}) = \Theta(n)$ and $R(\mathsf{DISJ}) = \Theta(n)$ (Rao and Yehudayoff 2020). We claim $R(\mathsf{TRAV}_2) \geq R(\mathsf{DISJ})$, and more generally $\mathsf{TRAV}_2$ is at least as hard as $\mathsf{DISJ}$.*

*Proof.* **Disjointness gadget.** We construct a gadget that enforces disjointness at a single entry $i \in [n]$. The gadget is shown in Figure **??**. There, let $x_i$ and $y_i$ be indicator functions whether $i \in S_A$ and $i \in S_B$ respectively. Then:

1. The top branch corresponds to when $i \in S_A$ but $i \notin S_B$.
2. The middle branch corresponds to when $i \notin S_A$ and $i \notin S_B$.
3. The bottom branch corresponds to when $i \in S_B$ but $i \notin S_A$.

Give $L$ and $R$ to $G_A$ and $G_B$. Then give all vertices labelled $x_i = 1$ to $G_A$ if this is true, else give all vertices labelled $x_i = 0$. Do the same with $y$ and $G_B$. In other words, there is no path $L$ to $R$ if $i \in X \cap Y$, but there is a path in all other cases.

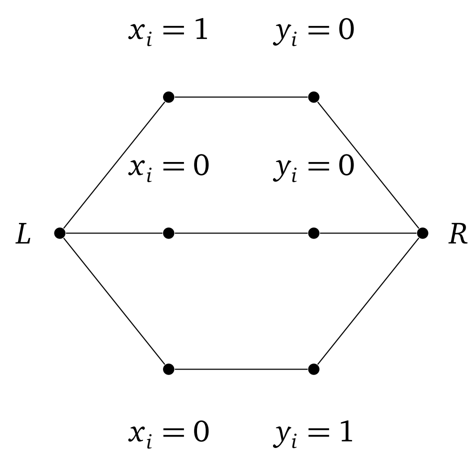


Figure 6: Disjoitness gadget. A path $L$ to $R$ only exists if either $x_i = 0$ or $y_i = 0$.

**Reduction to our graph problem.** We construct a graph $G$ by chaining one copy of this gadget for each index $i \in [n]$ between a global start vertex $v$ and end vertex $w$. Construct $G_A$ by giving all vertices at the start and end of every disjointness gadget (i.e. all $L$ and $R$), and vertices labelled $x_i = 1$ if it is true, else $x_i = 0$. Do similar for $G_B$.

In the resulting graph, there exists a path from $v$ to $w$ if and only if all disjointness gadgets are traversable, which occurs if and only if $X \cap Y = \emptyset$.

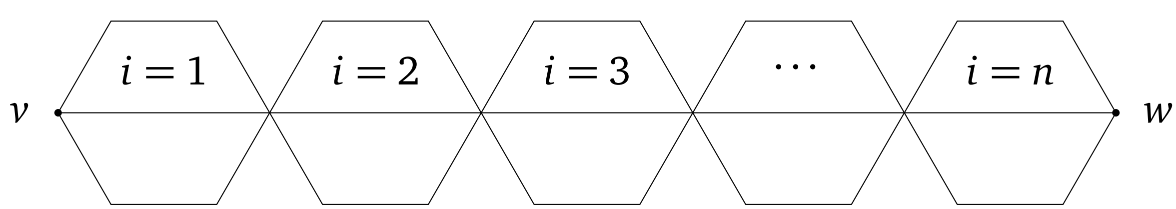


Figure 7: Disjointness gadgets chained together.

We conclude by noting that $|V(G)| = \Theta(n)$ and $|E(G)| = \Theta(n)$. As $|x| = |y| = \Theta(n)$, this completes the reduction and $\mathsf{TRAV}_2$ is at least as hard as $\mathsf{DISJ}$. $\square$

Recall that in Claim **??**, we showed that $D(\mathsf{TRAV}_2) = \Theta(n)$. With disjointness, we show that

$R(\mathsf{TRAV}_2) = \Theta(n)$ too. This is because Claim **??** gives $R(\mathsf{TRAV}_2) = \Omega(n)$, since $R(\mathsf{DISJ}) = \Theta(n)$ (Rao and Yehudayoff 2020). Then, the trivial protocol where Alice sends her entire $G_A$ to Bob in $\Theta(n)$ bits works even in the public coin randomness model. Together this gives $R(\mathsf{TRAV}_2) = \Theta(n)$. In other words, we cannot do much better even with bounded error.

## 2.3 Specialized Two-Party Bounds

The previous proofs show that $\mathsf{TRAV}_2$ is maximally complicated in terms of deterministic communication complexity and public randomized communication complexity. A natural follow up question is to ask if we can do better. That is, given some additional constraints, can we solve the problem with less communication asymptotically?

**Definition 2.** *Define the* $\mathsf{CTRAV}_2$ *problem as a* $\mathsf{TRAV}_2$ *problem where* $G_A$ *and* $G_B$ *are guaranteed to be complete and connected subgraphs of* $G$.

**Claim 4.** $\mathsf{CTRAV}_2$ *is at least as hard as* $\mathsf{DISJ}$. *Thus,* $D(\mathsf{CTRAV}_2) = \Theta(n)$ *and* $R(\mathsf{CTRAV}_2) = \Theta(n)$.

*Proof.* If the start $v$ or end $w$ is in neither $G_A$ nor $G_B$ or both in one of $G_A$ or $G_B$, the problem can be resolved in $O(1)$ communication. So suppose $v \in G_A$ and $w \in G_B$.

Since $G_A$ and $G_B$ are complete, a valid path occurs if and only if $V(G_A) \cap V(G_B) \neq \emptyset$. Then, by a similar argument to Claim **??**, this problem is at least as hard as $\mathsf{DISJ}$. More explicitly, let $x, y$ be inputs to $\mathsf{DISJ}$. We convert $x, y$ into subgraphs $G_A, G_B$ of a complete graph $G$ with $n$ vertices. This conversion is by including vertex $i$ in $G_A$ if $x_i = 1$, and not including it if $x_i = 0$. Do the same fot $G_B$ and $y$.

If $\mathsf{CTRAV}_2(G_A, G_B) = 1$, then $V(G_A) \cap V(G_B) \neq \emptyset$. So $x \cap y \neq \emptyset$ and $\mathsf{DISJ}(x, y) = 0$. Similarly, if $\mathsf{CTRAV}_2(G_A, G_B) = 0$, then $V(G_A) \cap V(G_B) = \emptyset$. Thus $x \cap y = \emptyset$ and $\mathsf{DISJ}(x, y) = 1$. We note that $|V(G_A)| = |V(G_B)| = |x| = |y| = \Theta(n)$. This completes the reduction. □

Informally, this tells us that the local structure of $G_A$ and $G_B$ alone is not a significant source of communication difficulty. That is, even when $G_A$ and $G_B$ are complete graphs, we do not immediately get better bounds and the problem does not become significantly easier. Since $G_A$ and $G_B$ are complete, we can say that these subgraphs have the maximum number of edges. We now consider a sparser case:

**Definition 3.** *Define* $\mathsf{STRAV}_2$ *as a* $\mathsf{TRAV}_2$ *problem where* $E(G) = O(n)$, $E(G_A) = O(n)$, *and* $E(G_B) = O(n)$. *In other words, the graph* $G$ *is sparse and* $G_A$, $G_B$ *are sparse too.*

**Claim 5.** $\mathsf{STRAV}_2$ *is at least as hard as* $\mathsf{DISJ}$. *Thus,* $D(\mathsf{STRAV}_2) = \Theta(n)$ *and* $R(\mathsf{STRAV}_2) = \Theta(n)$.

*Proof.* We reuse the reduction from $\mathsf{DISJ}$ to $\mathsf{TRAV}_2$ in Claim **??**. In that construction, $G$ is a chain of $n$ disjointness gadgets, so $|E(G)| = \Theta(n)$. Moreover, each player's subgraph consists of the gadget boundary vertices and one branch choice per gadget, which also gives $|E(G_A)| = \Theta(n)$ and $|E(G_B)| = \Theta(n)$.

Hence every instance produced by that reduction already satisfies the sparsity constraints in the definition of $\mathsf{STRAV}_2$. The reduction still preserves correctness: $\mathsf{DISJ}(x, y) = 1$ iff there is a $v$-$w$ path in $G_A \cup G_B$. Therefore $\mathsf{STRAV}_2$ is at least as hard as $\mathsf{DISJ}$. □

Informally, this tells us that weak edge sparsity of $G$, $G_A$, and $G_B$ does not add nor remove significant communication cost. This may lead us to believe that the local structure of $G$, $G_A$, and $G_B$, namely, how sparse or dense the edges corresponding to the owned vertices are, do not have a significant impact on the deterministic and pubic randomized communication complexity of the $\mathsf{TRAV}_2$ problem.

# 3 The Multi-Party Traversal Problem

In section **??**, we considered $\mathsf{TRAV}_2$, the communication game where both Alice and Bob computed the minimum length path from $v$ to $w$ using only edges in their subgraphs $G_A$ and $G_B$. Here, we generalize the problem to $\mathsf{TRAV}_k$, with $k$ players.

The rest of this section is under construction!

# 4 Closing Remarks

In this paper, we explored the communication complexity of multiparty graph traversal. By exploring embeddings in equality an disjointness we have been able to prove that our graph traversal problem is maximally hard. Our findings showcase our exploration in embedding known problems into novel problems.

Future research that expanding upon our findings could include various iterations of our problem. It could be interesting to explore different graph structures with properties that would allow for more efficient communication. It would also be interesting to investigate tighter bounds or proof techniques that may indicate stronger results.


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# Appendices

## A Multi-Party Graph Traversal Case Studies

Here, we survey a variety of real-world case studies where communication was a major bottleneck. Note that this section is a work in progress, especially in terms of formal citations.

**Air Transportation** Large-scale transportation and infrastructure disruptions create a natural model for multi-party graph traversal. Consider the 2010 European volcanic ash shutdown: air-navigation service providers, airports, and airlines each control disjoint portions of airspace and schedules[1]. Rerouting passengers is a shortest-path problem in a layered graph whose feasible vertices and edges change over time. No party can broadcast its full real-time state, both for operational and privacy reasons, so route feasibility must be computed with limited information exchange.

**Urban Transit** Similar graph-traversal constraints appear in urban transit shutdowns and phased restorations, where distinct agencies own stations and tunnels. After Hurricane Sandy, service availability evolved over weeks, and agencies coordinated alternate routes under partial knowledge of what infrastructure was open[2]. Highway detours after the I-95 bridge collapse in Philadelphia illustrate the same phenomenon in a road network: different operators own highway segments and toll roads, and a single failed cut edge forces traffic onto alternative paths with asymmetric knowledge of capacities[3].

**Maritime Logistics** The maritime and logistics examples are equally instructive. The 2021 Suez Canal blockage forced carriers to reroute around the Cape of Good Hope or wait, a decision that depends on schedule and capacity information held privately by shipping lines and port operators[4]. When the Baltimore Key Bridge collapsed, access to the port was blocked and cargo was diverted to alternate ports, again requiring coordination across a sea–port–land graph with private operational constraints[5].

[1] Prata and Rose 2015.
[2] Donovan and Work 2017.
[3] Pennsylvania Department of Transportation 2026.
[4] Wan et al. 2023.
[5] PBS NewsHour 2024.

**Infrastructure** Communication bottlenecks become even sharper in critical infrastructure networks. The Colonial Pipeline cyber disruption similarly forced fuel logistics to reroute through alternate depots and terminals with incomplete visibility[6]. And the 2021 Meta backbone withdrawal[7] and the 2016 Dyn DNS attack[8] show that digital services rely on reachability in routing and dependency graphs, where policy and security constraints limit information sharing across administrative domains.

These cases motivate a theoretical study of multi-party graph traversal: agents must collaboratively decide reachability and shortest paths while their private subgraphs, capacities, or weights are not fully shareable. The fundamental question is how much communication is necessary, and how randomized protocols, sketches, or partial disclosures can reduce the burden while preserving correctness.

# B Weekly Contributions

We summarize each member's weekly contributions below:

Table A 1: Weekly contributions by team member.

| Week | Ryan Batubara | Ivy Hawks | Darren Ho | Ciro Zhang |
|---|---|---|---|---|
| 6 | Work on formally writing up proofs for the trivial and other problems for our final report. | Work on formally writing up the proofs for the two party graph traversal problems. | Work on formally writing up the problem statement and motivation for the problem for our final report. | Work on formally writing up the introduction/motivation for communication complexity for our final report. |
| 5 | Help Ivy make the website. Write Github actions to publish to Github Pages. Make subposts components. Wrote CSS. | Start a Github pages website with our latest findings. Convert latex to markdown using Pandoc. | Write up formalized equality reduction in the report. Include the updated simplified procedure from the Professor. | Integrate all notes so far into the report. Fix typos etc in the other people who are writing the report. Give formal definitions, research sections. |

[6] Congressional Research Service 2021.
[7] Cloudflare Blog 2021.
[8] Antonakakis et al. 2017.

| Week | Ryan Batubara | Ivy Hawks | Darren Ho | Ciro Zhang |
|---|---|---|---|---|
| 4 | Introduction section for report, defining communication complexity and various applications of problems. | Look into embedding 3SAT into graph problems. Useful to see techniques to embed hard problems into our problem, especially if we want to prove it is in NP. | Look deeper into random protocols. In particular, develop a bank of random protocols that we can embed into our problem. | Draw figures of our problem and embeddings in LaTeX for the report and poster. |
| 3 | Looked at extreme case examples in real-life transport networks, e.g. failure of a node, failure of a "central hub" e.g. in the case on natural disasters, etc. | Wrote down problem statement and preliminary problem solution in latex using hypergraph. | Presented what random protocols are and several different random protocols, namely private and public randomness. | Wrote notes on various variations of shortest path hypergraph problems. |
| 2 | Identified various problems that involved the readings that we have found so far. Determined multiparty communication networks with self-driving cars shows promise. Showed how set disjointness can be used to determine collisions with self-driving cars. Also identified a protocol to compute this. Formalized potential problem. | Presented two papers on communication complexity of graph properties to see if there is any information that could be applied to self driving cars. Also presented problems that we will begin working on, with identifying a class of problems where the communication complexity is $o(n)$ by using the blackboard or third party model. Formalized potential problem. | Presented two papers on communication complexity of graph problems. One of them was on determining graph cliques, which can be used to determine whether or not there exists a clique, or a complete network. The other was to show a potential idea on how we can use ideas from other fields of study to compare with communication complexity. Took notes. | Presented a referee model, which we deduced to be the same as the blackboard model. We determined that this idea does not really work for our use case. Identified a critical flaw within our idea that was addressed. Found that our problem was too trivial and by removing/increasing certain assumptions the results remained trivial. |

| Week | Ryan Batubara | Ivy Hawks | Darren Ho | Ciro Zhang |
|---|---|---|---|---|
| 1 | Drafted communication problems with multiparty graph traversal avoiding intersections. Attempted hidden matching problem communication complexity but determined that it would be trivial by having each person explore all nodes. | Drafted communication problems with multiparty weighted graph intersection. Determined that the majority of problems and solutions involved trivial solutions. | Drafted communication problems with regards to black hole problems where there is a node that eliminates all players. Determined that finding it would be trivial by only communicating after 2 or n moves. Recorded meeting notes. | Drafted communication problems where everyone knows about their distance from each other but not any information about the graph. Determined that it would be trivial. |

# C Updated Proposal

We take a graph, upon which there are players who wish to identify if the graph possesses some property. We wish to use this model to find the minimum communication required for each player to identify if this property exists. Each Player will receive some portion of the graph as their input–this can be either vertices, edges, or both. We mostly study reachability of vertices, in which the only paths considered valid are paths which use only the portions of the graph given to the players as inputs.

For motivation, please see Section **??** of the report.

## C.1 Project Goals

Sections **??**, **??**, **??**, and **??** formally define our problem statement and the full spectrum of problems we are considering for our project.

Clearly, section **??** is done. Much of the results for the remaining sections are also complete, and it only remains to formalize and type them up in LaTeX.

## C.2 Actionables

Therefore, this project can be summarized into three specific, actionables:

1. Finish writing this paper to outline the formal, technical communication complexity bounds for a given multi-agent traversal problem with certain constraints.
2. Finish porting the report to our website, whose link is available on the cover page of this report.
3. Give a poster presentation of the key results of the paper, with the intention of making the results applicable to a wider audience.

Therefore, our work will remain theoretically grounded while remaining applicable for the wider computing community.

## C.3 Deliverables

We have three main deliverables:

1. A paper presenting actionable item 1 and 2.
2. A research poster of our key results, for actionable 3.
3. A website containing the paper and poster information, for actionable 3.

To be more specific, the paper will take the form of a research paper, where we define the problems we are working on, prove bounds on these problems, and show how they model situations in the field of multi-agent graph traversal. The paper is our primary project output.

Our detailed weekly progress is shown in the previous appendix section. In the remaining time, we plan for the following actionables for each person:

Table A 2: Weekly plan

| Week | Ryan Batubara | Ivy Hawks | Darren Ho | Ciro Zhang |
|---|---|---|---|---|
| 7 | Finish the other problems section, with a focus on figures. | Finish the iterated multi-party section, with a focus on figures. | Make significant progress on the multi party section, with a focus on figures. | Make significant progress on the iterated multi-party section, with a focus on figures. |
| 8 | Finish the multi-party section. | Finish the iterated multi-party section. | Finish the website port. | Move figures to first draft of poster. |
| 9 | Polish paper and proofread website. | Polish website and proofread poster. | Polish poster and proofread paper. | Polish poster and proofread paper. |

Indeed, given our weekly contributions thus far we are on track to finishing the project with a week dedicated solely to polishing.